

Due on monday, November 14th, at the beginning of tutorial.

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This assignment is Questions 1 and 3 of the Autumn Repeat Exam of the 2015-2016 academic year. The exam paper is put up on the module webpage; you can of course download it from the library as well.

Please answer questions 1 and 3. The assignment marks will be half the marks listed in the exam.

Some instructions/comments below.

1. (a) Make sure you show explicitly how you are using the orthonormality of $\{|\phi_n\rangle\}$ in the calculation of the norm.
(b) You will have to use the fact that the $\{|\phi_n\rangle\}$ are energy eigenstates.
(c) This question is about measurement in quantum mechanics.
For guidance, you could try reading the first few sections of the wikipedia page on “Measurement in Quantum Mechanics”, and Sections 3.2 and 3.3 of Nash notes. There is also a paragraph summarizing measurement in the writeup “Essentials of QM”, linked on the webpage.
2. —
3. (a) More about measurement.
(b) This one needs some serious calculation. This semester we used the notation $\hat{x}$ and $\hat{p}$ for position and momentum operators. The exam uses X and P to mean the same things.
(c) Please look up Ehrenfest’s theorem, of which this is a special case.